# Table of Content

# Chapter 1: Operating System Overview

## 1.1 Introduction

An Operating System (OS) is an interface between a computer user and computer hardware. An operating system is a software which performs all the basic tasks like file management, memory management, process management, handling input and output, and controlling peripheral devices such as disk drives and printers. An operating system (OS) is system software that manages computer hardware, software resources, and provides common services for computer programs.

An operating system is software that enables applications to interact with a computer's hardware. The software that contains the core components of the operating system is called the kernel.

The primary purposes of an Operating System are to enable applications to interact with a computer's hardware and to manage a system's hardware and software resources.

## 1.2 Evolution of Operating System

- Stanford Research Institute developed the oN-Line System (NLS) in the late 1960s, which was the first operating system that resembled the desktop operating system we use today.
- Microsoft bought QDOS (Quick and Dirty Operating System) in 1981 and branded it as Microsoft Operating System (MS-DOS). As of 1994, Microsoft had stopped supporting MS-DOS.
- Unix was developed in the mid-1960s by the Massachusetts Institute of Technology, AT&T Bell Labs, and General Electric as a joint effort. Initially it was named MULTICS, which stands for Multiplexed Operating and Computing System.
- FreeBSD is also a popular UNIX derivative, originating from the BSD project at Berkeley. All modern Macintosh computers run a modified version of FreeBSD (OS X).

- Windows 95 is a consumer-oriented graphical user interface-based operating system built on top of MS-DOS. It was released on August 24, 1995 by Microsoft as part of its Windows 9x family of operating systems.
- Solaris is a proprietary Unix operating system originally developed by Sun Microsystems in 1991. After the Sun acquisition by Oracle in 2010 it was renamed Oracle Solaris.

## 1.3 Function of Operating System

1. **Security**

The operating system uses the password to protect user data and similar other techniques. It also prevents unauthorized access to programs and user data. Operating systems are large and complex programs, and often security issues will be found. Often a virus or worm will take advantage of these bugs to access resources it should not be allowed to, such as your files or network connection; to fight them you must install patches or updates provided by your operating system vendor.

2. **Multitasking**

The operating system is responsible for resource management within the system. Many tasks will be competing for the resources of the system as it runs, including processor time, memory, disk and user input. The job of the operating system is to arbitrate these resources to the multiple tasks and allow them access in an orderly fashion.

3. **Error detecting aids**

Operating system keeps tracks of time and resources used by various tasks and users, this information can be used to track resource usage for a particular user or group of users.

4. **Control overall system performance**

As the operating system provides so many services to the computer, its performance is critical. Many parts of the operating system run extremely frequently, so even an overhead of just a few processor cycles can add up to a big decrease in overall system performance.

The operating system needs to exploit the features of the underlying hardware to make sure it is getting the best possible performance for the operations, and consequently systems programmers need to understand the intimate details of the architecture they are building for.

5. **File Management**

A file system is organized into directories for efficient or easy navigation and usage. These directories may contain other directories and other files. An Operating System carries out the file management activities. It keeps track of where information is stored, user access settings and status of every file and more. These facilities are collectively known as the file system.

# Chapter 2: Structure Of Operating System

The structure of an operating system defines how its components are organized and interact with each other to manage computer resources effectively. The operating system can be implemented with the help of various structures. The structure of the OS depends mainly on how the various standard components of the operating system are interconnected and merge into the kernel.

## 2.1 Windows Operating System

### 2.1.1 Introduction

Windows operating system is a computer program that manages all computer resources and provides services to applications that run on top of it. It features the first graphical user interface (GUI) for IBM-compatible PCs, the Windows OS soon dominated the PC market. Approximately 90 percent of PCs run some version of Windows.

This operating system was developed by Microsoft and released in 1985 under the name Windows 1.0. Since then, the Windows operating system has continued to evolve and become one of the most popular operating systems in the world. The Windows operating system is designed to run on various types of hardware, including desktops, laptops, servers, and mobile devices.

Windows uses a graphical user interface (GUI) that allows users to interact with the computer through icons, buttons, and visual menus, rather than using text commands like other operating systems. Windows also has many features such as multitasking capability, which allows several applications to run simultaneously, as well as plug-and-play capability that makes it easy for users to connect additional devices such as printers, scanners, and cameras.

### 2.1.2 Functions of the Windows Operating System

**I. Managing Computer Resources:**

The primary function of the Windows operating system is to manage and organize computer resources such as CPU, RAM, and hard disk. The Windows operating system will complete various tasks such as opening applications, accessing the internet, and printing documents using these resources.

**II. Providing an Interface:**

The Windows operating system provides a Graphical User Interface (GUI) that allows users to access and use various applications and programs easily. This interface allows users to select menus, click icons, and navigate various applications easily.

**III. Providing Compatibility:**

The Windows operating system is designed to support various hardware and software devices, making it easier for users to install and use different applications and programs on their computers or laptops. Windows also provides the ability to run programs and applications designed for different versions of the Windows operating system.

**IV. Facilitating Security:**

The Windows operating system provides various security features such as anti-virus, anti-malware, and firewall that help protect computers or laptops from virus and malware attacks. Windows also provides tools to configure network security and access control to protect user's important data.

**V. Managing File Management:**

Windows also facilitates file management such as data storage, access rights configuration, and file searching. Windows provides various tools to help users organize and store data and files on their computers or laptops.

### 2.1.3 Advantage of Windows Operating System

**I. Wide Compatibility**

Windows OS is widely known to have a diverse compatibility of numerous hardware and software. This allows it to support many device manufacturers, giving users significant flexibility in hardware configuration. Windows also supports thousands of applications, including productivity and gaming software. Its versatility provides support for personal, educational, and professional bases. It also has enhanced versatility through its ability to integrate with peripherals such as printers, scanners, and external drives.

**II. Convenience**

All forms of Microsoft Windows have something regular in it which makes it clients simple to move starting with one form then onto the next. Windows 7 clients have no trouble in moving to Windows 10 in light of the fact that a large portion of the highlights of Windows 10 is equivalent to Windows 7. The UI of windows is additionally simple to use than UNIX and MAC.

**III. Programming support**

Windows stage is most appropriate for game and programming engineers. Windows have a huge number crowd so designers want to make utilities, games, and programming for windows OS. Linux clients can't make windows applications so it is smarter to utilize windows for creating applications.

**IV. Ease of use:**

All versions of Microsoft Windows have something common in it which makes users easy to shift from one version to another. Windows 7 users have no difficulty in migrating to Windows 10 because most of the features of Windows 10 are the same as windows 7. The user interface of windows is also easy to use than UNIX and MAC.

**V. User-Friendly Interface**

Khmer642 — Windows has a very simple and easy-to-use GUI. Elements such as the Start menu, taskbar, and desktop icons allow quick access to commonly used applications and settings. The OS also offers customization options, so users can tailor their experience. Microsoft has used over this time to fine-tune the interface so that it works seamlessly, making it suitable for individual users and businesses alike.

### 2.2.4 Disadvantage of Windows Operating System

**I. Security Concerns**

Windows operating system is relatively unsafe against security threats. Therefore, hackers can easily penetrate into the system by breaking its security layers. So, users of windows operating system are to be highly rely on anti-virus software to protect their data from malicious attacks. For accessing service of these anti-virus software, one has to pay a subscription fee. Another major issue of windows operating system is that one has to update the operating system regularly to keep it up to date with security patches.

### II. Complex Licensing and Activation

Windows licensing and activation methods can be confusing, especially for businesses that are overseeing multiple devices. Users must ensure adherence to Microsoft's licensing terms, a costly and complex process. Failure to comply will cause restricted access or complications in legal actions. Moreover, the licensing is often not transferable between devices, making hardware upgrades or replacements difficult.

### III. Paid Software

The application software run on windows operating system are paid. For example, we want to do graphic designing, we have purchase Photoshop or any other graphics software, for word processing we have to purchase MS-Word, and so on.

### IV. Expensive

To use Windows operating system on the computer, we have to purchase license from Microsoft, thus we cannot use windows operating system legally free. The cost of buying a copy of windows license is very high.

### V. Poor Technical Support

Windows customer support is very poor for most windows users. Windows support team provides technical support service only to some large organizations. Therefore, individual users have to look at some external resource and forums to resolve their technical problems.

## 2.2Linux Operating System

### 2.2.1 Introduction

Linux is an open-source operating system (OS) created by Linus Torvalds in 1991. Today, it has a massive user base, and is used in the world's 500 most powerful supercomputers. Users gravitate toward it for its versatility and security capabilities, among other reasons. The Linux kernel is maintained by a worldwide community of open-source enthusiasts and has hundreds of unique distros.

It relies heavily on the Linux kernel for its functionality. In an OS, the kernel is a computer program that allows users to control the system's hardware and software. In addition to the kernel, the Linux OS uses various components, such as system libraries and space utilities, but they all rely on the kernel to communicate and receive commands from users.

In many ways, Linux is similar to other operating systems such as Windows, macOS (formerly OS X), or iOS. Like other operating systems, Linux has a graphical interface, and the same types of software you are accustomed to, such as word processors, photo editors, video editors, and so on. In many cases, a software's creator may have made a Linux version of the same program you use on other systems.

But Linux also is different from other operating systems in many important ways. First, and perhaps most importantly, Linux is open-source software. The code used to create Linux is free and available to the public to view, edit, and—for users with the appropriate skills—to contribute to.

Linux is also different in that, although the core pieces of the Linux operating system are generally common, there are many distributions of Linux, which include different software options. This means that Linux is incredibly customizable, because not just applications, such as word processors and web browsers, can be swapped out. Linux users also can choose core components, such as which system displays graphics, and other user-interface components.

### 2.2.2 Functions of the Linux Operating System

**I. Portable Environment**

Linux software operates flawlessly on a variety of hardware platforms. Without the worry of incompatibility, individuals can use Linux operating system on any device. It runs the same way on both high-end and low-end hardware.

**II. Free and Open-Source**

Its source code is available for anybody to use and alter. Many developers collaborate in organizations to improve and strengthen Linux, and lots of developers constantly work on updating the Linux system.

**III. Graphical User Interface (GUI)**

It comes with Graphical User Interface (GUI) abilities in the same way you can with Windows. Similarly, users can install the programs, and the computer graphics will begin to work in the same way that Windows does.

**IV. Frequent New Updates**

Software updates are controlled by the users in Linux. Individuals have the option to pick and choose which updates are required, and there is a plethora of system updates accessible. These upgrades happen considerably more quickly than on other operating systems. Therefore, system upgrades can be deployed without difficulty.

**V. Extremely Flexible**

It is highly flexible, and a variety of desktop applications, embedded systems, and server applications can benefit from the same. It also offers a number of computer-specific limitation settings for admins to allow only essential components to get installed.

### 2.2.3 Advantage of Linux Operating System

**I. Open Source**

One of the main advantages of Linux is that it is open-source. Linux allows users to access, configure and modify the software freely compared to other operating systems. The collaborative nature of its development means that more eyes are there on the code, which can lead to faster bug fixes and improvements.

**II. Cost-Effectiveness**

Most Linux distributions are free to use. This can be a significant financial benefit, especially for businesses looking to minimize costs on software licenses. While some specialized versions may have fees, the vast majority are freely available without the hidden costs of proprietary software.

**III. Personalization**

The flexibility of Linux allows users to create a computing environment tailored to their specific needs. From lightweight desktop environments to full-fledged software stacks, the level of customization available is unmatched.

**IV. Security Features**

With its robust user permissions and inherent open-source model, Linux is generally considered more secure than many other operating systems. While no system is completely immune to security threats, Linux users often benefit from proactive and frequent security updates from the community.

**V. Performance and Stability**

Linux is known for its stability and performance. It can run on older hardware where newer versions of Windows might struggle. Additionally, Linux tends to experience fewer crashes, making it a reliable choice for servers and critical applications.

### 2.2.4 Disadvantage of Linux Operating System

**I. Learning Curve**

For users accustomed to Windows or macOS, switching to Linux may present a steep learning curve. The user interface, system commands, and overall usage can feel unfamiliar to those new to the platform.

**II. Hardware Compatibility**

While Linux supports a wide array of hardware, there may still be instances where specific drivers or components don’t work seamlessly. This can be particularly challenging for users with newer or less common hardware.

**III. Software Compatibility**

Although many well-known applications have Linux versions, some proprietary software, particularly specialized tools in design or gaming, may only be available on Windows. This can limit options for users who rely on certain software for their work.

**IV. Fragmentation**

Due to the many Linux distributions available, fragmentation can become an issue. This means that certain applications or software may not work across all Linux distributions, leading to confusion or additional troubleshooting.

**V. Lack of Commercial Support**

Linux lacks the same level of commercial support as competing operating systems. This can make it difficult for users to obtain assistance when operating system problems or issues arise. For instance, a user may be unable to locate a commercial support team that can assist them with Linux troubleshooting.

## 2.3Mobile Operating System

### 2.3.1 Introduction

A mobile operating system is an operating system that is specifically designed to run on mobile devices such as mobile phones, smartphones, PDAs, tablet computers and other digital mobile devices. A Mobile Operating System is a software interface designed to manage hardware components and assist users in utilizing their mobile devices, such as smartphones and tablets, similar to how personal computers were used in the past. Popular mobile operating systems such as android, Huawei, and iOS, etc.

Modern mobile operating systems combine the features of a personal computer operating system with touchscreen, cellular, Bluetooth, WIFI, GPS mobile navigation, camera, video camera, speech recognition, voice recorder, music player, Near field communication, personal digital assistant (PDA) and other features. Mobile OS is a mobile version of an operating system found on computers. Much like the Linux or windows operating system controls your desktop or laptop computer, a mobile operating system is the software platform on top of which other programs can run on mobile devices.

### 2.3.2 Features of the Mobile Operating System

**I. User Interface (UI):**

Touch inputs of Graphical User Interface (GUI) provided by mobile OS are optimized. This is where users can use touch gestures, in other words, swiping, tapping, and pinching, to interact with their gadgets.

**II. Multitasking:**

It helps in running of many apps at the same time but what is more we can quickly switch between them without any hindrance. Such offloading is for applications that are not currently used actively.

**III. Connectivity:**

It provides a variety of connections such as cellular, Wi-Fi, Bluetooth, NFC (Near Field Communication) and others to facilitate the communication of the device with other devices and networks.

IV. **Application Management:**

Is a platform that has its own app marketplace or store which the users utilize to browse, install, run and updates the applications exclusively for that platform.

**V. Resource Management:**

Efficiently allocates hardware resources like the CPU, RAM, and battery by achieving a balance between performance and battery life.

### 2.3.3 Advantage of Mobile Operating System

**I. User-Friendly Design**

Smartphones are designed to be simple and easy to use. Unlike computers, most people can operate them without special training. The touch screen, icons, and menus make navigation straightforward, even for beginners.

**II. Lightweight and Power Efficient**

Being small and lightweight, smartphones are highly portable. They also consume less power compared to computers, which allows them to run for longer on a single charge—something very important for mobile devices.

**III. Customization**

Smartphones can be customized to suit personal preferences. Users can change wallpapers, rearrange icons, or download themes. The operating system also updates itself automatically, ensuring the phone stays smooth, secure, and up to date with the latest features.

**IV. Applications**

A major advantage of smartphones is the wide range of apps available in app stores. Whether for productivity, education, social media, entertainment, or gaming, users can easily download apps to match their needs.

**V. Security Features**

Smartphones come with strong security measures such as passwords, PINs, and biometric options like fingerprint or face recognition. If a phone is lost or stolen, tracking apps can help locate it, and in some cases, the device can be locked or erased remotely to protect personal data.

### 2.3.4 Disadvantage of Operating System

#### I. Battery Drain

Most mobile operating systems allow apps to run in the background. While this improves multitasking, it also consumes a lot of power, making smartphones drain their battery quickly compared to other devices.

#### II. Performance Limitations on Low-End Devices

Mobile OS require significant resources to run smoothly. In low-specification smartphones, the operating system and apps often cause slow performance, lagging, or overheating, reducing the overall usability of the device.

#### III. Security Concerns

Mobile operating systems are vulnerable to security risks. Malicious apps, data theft, and viruses can affect both Android and iOS. Users may unknowingly download harmful apps, putting personal data such as photos, messages, and banking details at risk.

#### IV. Excessive Advertisements in Apps

Many applications available on mobile platforms are free, but they display frequent advertisements. These ads interrupt the user experience, reduce productivity, and sometimes even mislead users into downloading unnecessary apps.

#### V. Software Update Limitations

Not all devices receive timely software updates. Older smartphones often stop receiving updates, leaving users without access to the latest features or security improvements. To get the newest version of the operating system, users usually need to buy newer and more expensive devices.

## ]2.4 Comparative Summary Table

| Feature | Windows OS | Linux OS | Mobile OS |
|---|---|---|---|
| **Primary Use** | Personal computers, desktops, laptops, and business workstations with wide hardware compatibility. | Servers, supercomputers, desktops (favored by tech enthusiasts and embedded systems). | Smartphones, tablets, and also powers wearables, IoT devices, and automotive systems. |
| **Target Audience** | General-purpose users needing ease of use, compatibility, and productivity. | Tech-savvy users, developers, system administrators seeking flexibility and control. | Primarily end-users focused on ease of use and mobility; apps are designed for different screen sizes. |
| **User Interface** | GUI with desktop icons, taskbar, Start menu, and intuitive navigation. | GUI + CLI (Bash) with customization and full control. | Touch-based interface with gestures and mobile-friendly navigation. |
| **App Ecosystem** | Broad, including productivity, gaming, enterprise tools. | Smaller ecosystem; focused on server/technical tools, but growing desktop apps. | Massive app ecosystem with millions of apps; restricted by app store policies. |
| **Hardware Compatibility** | Wide support provided by Microsoft and hardware vendors. | High compatibility, but some custom devices may need extra drivers/config. | Mobile devices tied to specific hardware; limited to supported devices only. |
| **Security** | Frequent updates, built-in antivirus (Windows Defender). Vulnerable to malware. | Strong security, permission management, open- | Varies: Android more open (higher risk), iOS closed |

| | | | |
|---|---|---|---|
| | | source community patches. | (tighter control, higher security). |
| **Stability** | Generally stable, occasional issues with updates and older hardware. | Very stable, servers can run for years without crashes. | Stable for mobile use, but updates may affect compatibility. |
| **Open Source** | Proprietary (source code closed). | Open-source (GNU GPL, free to modify and redistribute). | iOS is proprietary; Android is open-source (AOSP) but often bundled with proprietary vendor apps. |
| **Cost** | Paid license (Windows Home, Pro, etc.). | Free and open-source (enterprise versions may require paid support). | Android is free (open-source); iOS comes bundled with Apple devices. |